



ATTRACTOR BUBBLING IN COUPLED HYPERCHAOTIC OSCILLATORS

JONATHAN N. BLAKELY and DANIEL J. GAUTHIER

*Department of Physics and Center for Nonlinear and Complex Systems,
Duke University, Box 90305, Durham, North Carolina 27708, USA*

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We investigate experimentally attractor bubbling in a system of two coupled hyperchaotic electronic circuits. The degree of synchronization over a range of coupling strengths for two different coupling schemes is measured to identify bubbling. The circuits display regimes of both attractor bubbling and high-quality synchronization. For the coupling scheme where high-quality synchronization is observed, the transition to bubbling is “soft” and its scaling with coupling strength near the transition point does not fit into the known categories of transition types. We also compare the observed behavior to several proposed criteria for estimating the regime of high-quality synchronization. It is found that none of these methods is completely satisfactory for predicting accurately the regimes of attractor bubbling and high-quality synchronization.

1. Introduction

One hallmark of chaos is that the trajectories of two identical chaotic systems starting from very close initial conditions diverge exponentially. It is now known that this divergence can be overcome by an appropriate coupling between the systems, thereby synchronizing their evolution [Fujisaka & Yamada, 1983; Pecora & Carroll, 1990]. In the past decade, the study of synchronized low-dimensional chaotic oscillators has inspired several possible applications including secure communication [Pecora *et al.*, 1997], dynamics-based computation [Sinha & Ditto, 1998], and schemes for controlling complex spatiotemporal dynamics. More recently, it has become clear that such applications require a better understanding of the dynamics of coupled higher dimensional systems displaying hyperchaos (dynamics characterized by multiple positive Lyapunov exponents). For example, schemes for secure communication should be based on synchronized hyperchaotic oscillators since low-dimensional chaos is not complex enough to securely mask information [Perez & Cerdeira, 1995; Short, 1997].

One fundamental issue in synchronizing hyperchaotic systems is the number of distinct coupling signals necessary to achieve synchronization. Early on, it was conjectured that the number of coupling signals must be greater than or equal to the number of positive Lyapunov exponents [Pyragas, 1993]. This conjecture has since proven to be untrue. Theoretical studies of several hyperchaotic systems show that it is possible to obtain asymptotic stability of the synchronized state using only a single coupling signal [Peng *et al.*, 1996; Tamaševičius & Cenys, 1997; Wang & Wang, 1998; Brucoli *et al.*, 1998]. Synchronized hyperchaos using such a scheme has been demonstrated experimentally in electronic circuits [Tamaševičius *et al.*, 1996, 1997b; Johnson *et al.*, 1998], erbium-doped fiber ring lasers [Van Wiggeren & Roy, 1998], and delayed feedback diode lasers [Goedgebuer *et al.*, 1998].

These experiments demonstrate that hyperchaos can be synchronized, but they do not report whether synchronization can be obtained over a wide range of coupling schemes and coupling strengths. It is important to determine the full

range of synchronization behavior to thoroughly test our theoretical understanding of this phenomenon. Large discrepancies between theory and experiment were found in studies of coupled low-dimensional systems [Ashwin *et al.*, 1994; Heagy *et al.*, 1995; Gauthier & Bienfang, 1996; Rulkov & Sushchik, 1997]. The source of these discrepancies is attractor bubbling, a new type of dynamical behavior where a small amount of noise or mismatch between oscillators [Pikovsky & Grassberger, 1991] gives rise to large intermittent bursts of desynchronization.

Attractor bubbling has yet to be investigated experimentally in systems of coupled hyperchaotic oscillators, although Ahlers *et al.* [1998] observed bubbling in a hyperchaotic numerical model of coupled diode lasers. None of the experiments cited above report the close comparison of theory and experiment over a range of coupling schemes and coupling strengths necessary to characterize attractor bubbling. Thus, important questions remain unanswered. Is the bubbling in hyperchaotic oscillators more severe than in low-dimensional systems given the greater complexity of hyperchaos? Can experiments on high-dimensional systems give additional guidance on the formulation of a method for predicting attractor bubbling?

In this paper, we address these questions through an experimental investigation of attractor bubbling in a system of coupled hyperchaotic electronic oscillators. We observe the degree of synchronization over a range of coupling strengths and coupling schemes in order to determine the prevalence of attractor bubbling. Our results show large regimes of both attractor bubbling and high-quality (bubble-free) synchronization. We also develop a precise mathematical model of our experimental system in order to compare our observations to several proposed methods for estimating the range of synchronization. We find that none of these methods is completely satisfactory for predicting accurately the regimes of attractor bubbling and high-quality synchronization.

This paper is structured as follows. In Sec. 2, we briefly review synchronization and attractor bubbling in low-dimensional systems. Next, we describe the experiments performed to investigate attractor bubbling in a system of two coupled hyperchaotic oscillators in Sec. 3 and present the results in Sec. 4. In Sec. 5, we apply several different criteria used to predict synchronization to our system and compare them to the

experimental results. Finally, in Sec. 6, we summarize our conclusions and discuss the implications of our results for using synchronized hyperchaos in secure communication applications.

2. Synchronization and Attractor Bubbling

In this section, we review the fundamental concepts underlying synchronization of chaotic oscillators, using a geometric description of this phenomenon. This sets the stage for discussing the previous research on attractor bubbling in systems of coupled low-dimensional oscillators. For concreteness, we limit our discussion to the case of two one-way coupled oscillators.

The dynamics of a coupled system of two oscillators can be expressed succinctly as

$$\frac{d\mathbf{x}_m}{dt} = \mathbf{F}[\mathbf{x}_m], \quad (1a)$$

$$\frac{d\mathbf{x}_s}{dt} = \mathbf{F}[\mathbf{x}_s] - c\mathbf{K}(\mathbf{x}_m - \mathbf{x}_s), \quad (1b)$$

where $\mathbf{x}_m$ ($\mathbf{x}_s$) denotes the position in the n -dimensional phase space of the master (slave) oscillator, $\mathbf{F}$ is the vector field governing the flow of a single oscillator, $\mathbf{K}$ is a $n \times n$ coupling matrix, and c is the scalar coupling strength. Synchronization of the oscillators is defined by $\mathbf{x}_s(t) = \mathbf{x}_m(t)$ and hence the evolution takes place on an n -dimensional hyperplane, called the *synchronization manifold*, within the full $2n$ -dimensional phase space. We introduce new coordinates specifying the dynamics within and transverse to the synchronization manifold as

$$\mathbf{x}_{\parallel} = (\mathbf{x}_m + \mathbf{x}_s)/2, \quad (2a)$$

$$\mathbf{x}_{\perp} = (\mathbf{x}_m - \mathbf{x}_s)/2, \quad (2b)$$

respectively. In this basis, the synchronization manifold is defined by $\mathbf{x}_{\perp}(t) = 0$.

According to Fujisaka and Yamada [1983], the synchronization manifold is asymptotically stable if and only if the largest *transverse Lyapunov exponent* is negative. These exponents are the average rates of exponential expansion or contraction in directions transverse to the synchronization manifold. They are determined from the solution of the variational equation

$$\frac{d\delta\mathbf{x}_{\perp}}{dt} = \{\mathbf{D}\mathbf{F}[\mathbf{x}_m(t)] - c\mathbf{K}\}\delta\mathbf{x}_{\perp}, \quad (3)$$

obtained by linearizing Eq. (1) about $\mathbf{x}_\perp = 0$, where $\delta\mathbf{x}_\perp$ is a small perturbation away from the synchronization manifold, and $D\mathbf{F}[\mathbf{x}_m(t)]$ is the Jacobian of the vector field $\mathbf{F}$ evaluated on the attractor of the master oscillator.

In the absence of noise and mismatch of the oscillator parameters, the stability of the synchronization manifold determines whether or not synchronization will occur. Fujisaka and Yamada apparently thought that their result remains valid when the oscillators are subject to small noise or are not perfectly matched. They described how to determine experimentally the largest Lyapunov exponent of an oscillator by observing the transition to synchronization as the coupling strength is varied. In just such an experiment, Schuster *et al.* [1986] determined the largest Lyapunov exponent of an oscillator by coupling it to a near-identical oscillator and observing the coupling strength at which the average value of $|\mathbf{x}_\perp|$ became small. Clearly, this method relies on the coincidence of the coupling strength at which synchronization becomes unstable and the coupling strength at which synchronization is lost experimentally. Many other researchers have assumed that the *asymptotic* stability of the synchronization manifold is a good predictor of synchronization in real physical systems despite the presence of noise and parameter mismatch. For example, a demonstration that the largest Lyapunov exponent is negative is presented as proof that synchronization of chaotic oscillators will occur (see e.g. [Pyragas, 1993; Peng *et al.*, 1996; Zonghua & Shigang, 1997]).

However, several researchers have found that this synchronization criterion is inconsistent with experimental results [Ashwin *et al.*, 1994; Heagy *et al.*, 1995; Gauthier & Bienfang, 1996; Rulkov & Sushchik, 1997]. In some synchronization experiments where all transverse Lyapunov exponents are negative, long intervals of synchronization are interrupted irregularly by large (comparable to the size of the attractor), brief desynchronization events. This behavior has been called *attractor bubbling* [Ashwin *et al.*, 1994]. Ashwin *et al.* [1994, 1996] pointed out that the synchronization manifold may contain unstable sets, such as UPO's, characterized by positive transverse Lyapunov exponents, even though the transverse exponents characterizing the chaotic attractor are all negative. Near such sets, the manifold is locally repelling so a small perturbation arising from experimental noise will result in a brief excursion away from the synchronization

manifold. These desynchronization events recur because the chaotic evolution of the system brings it into the neighborhood of the repelling set an infinite number of times. This explanation implies that the transition from bubble-free synchronization to bubbling as the coupling strength is varied occurs when a set first becomes transversely unstable. Building on these ideas, Venkataramani *et al.* [1996a, 1996b] identified generic types of transitions from synchronization to bubbling by considering the behavior of coupled 2-D maps and electronic oscillators.

Several researchers have attempted to characterize attractor bubbling in experiments on coupled low-dimensional chaotic circuits. Heagy *et al.* [1995] located experimentally several unstable periodic orbits (UPO) embedded in the attractor of a chaotic circuit and then used the UPO's to drive a slave oscillator. They determined the critical coupling strength required for synchronizing the slave oscillator to each UPO and found that it varied from orbit to orbit. In addition, some UPO's required a coupling strength larger than that needed to make the largest transverse Lyapunov exponent negative. In a separate experiment, Gauthier and Bienfang [1996] observed that the loss of synchronization of a system of coupled chaotic circuits coincided with the first loss of transverse stability by a UPO as the coupling strength was decreased. In another study of coupled circuits, Rulkov and Sushchik [1997] found "disruptive" regions on the synchronization manifold that were characterized by large positive local transverse exponents. These disruptive regions were correlated with desynchronization events and associated with UPO's embedded in the attractor. Recently, bubbling-like behavior has been observed in coupled periodic oscillators where UPO's play no role [Gauthier, 1998; Blakely *et al.*, 1999]

Attractor bubbling clouds the question of whether two oscillators are synchronized in an experiment. This is because, in the presence of bubbling, the oscillators are synchronized on average (the time averaged value of $|\mathbf{x}_\perp(t)|$ is very small) but large desynchronization events still occur intermittently (the maximum value of $|\mathbf{x}_\perp(t)|$ is large). To emphasize this distinction, Gauthier and Bienfang [1996] referred to synchronization as *high-quality* if $|\mathbf{x}_\perp(t)| < \varepsilon$ for all time $t > 0$, where ε is a length scale small in comparison to the characteristic size of the chaotic attractor. In this definition, the metric, the value of ε , and the norm used for determining $|\mathbf{x}_\perp(t)|$ are arbitrary. However, in many

circumstances there may be a physically relevant choice of metric, ε , and norm.

In light of the number experiments in which attractor bubbling has been observed, it appears that bubbling is a typical behavior of coupled low-dimensional chaotic oscillators. Hyperchaotic dynamics is significantly more complex and hence one might expect a higher prevalence of attractor bubbling as the complexity increases. In the next section, we describe our experimental system for investigating attractor bubbling in coupled hyperchaotic oscillators.

3. Experimental Apparatus

Our experimental system consists of a pair of well-matched coupled hyperchaotic electronic circuits based on the design of Tamaševičius *et al.* [1997a]. Electronic circuits are ideal for studying attractor bubbling because several coupling schemes can be implemented and the coupling strength varied in order to determine the prevalence of bubbling, and accurate models can be developed for comparing experimental observations to theoretical predictions.

A single circuit, shown schematically in Fig. 1, consists of inductors $L_1 = 250$ mH and $L_2 = 97$ mH, capacitors $C_1 = 198$ nF and $C_2 = 60$ nF, a 1N914 diode, and a negative resistor $-R$ implemented using an Analog Devices OP-42 op amp. The dynamical evolution of the circuit is described by the set of equations

$$\frac{dv_1}{dt} = av_1 - i_1 - g(v_1 - v_2), \quad (4a)$$

$$\frac{dv_2}{dt} = -bi_2 + bg(v_1 - v_2), \quad (4b)$$

$$\frac{di_1}{dt} = v_1 - hi_1, \quad (4c)$$

$$\frac{di_2}{dt} = dv_2 - ei_2, \quad (4d)$$

where v_1 and v_2 represent the voltage drop across the capacitors. The variables i_1 and i_2 represent the current flowing through the inductors scaled by the characteristic resistance $\rho = \sqrt{L_1/C_1}$ so that they have dimensions of voltage. The other parameters are given by $a = \rho/R$, $b = C_1/C_2$, $d = L_1/L_2$ and time is normalized to $\tau = \sqrt{L_1 C_1} = 0.2$ ms. Also, $h = r_1/\rho$ and $e = dr_2/\rho$, where $r_1 = 96 \Omega$ and $r_2 = 57 \Omega$ are the DC resistances of L_1 and L_2 , respectively. The current flowing through the diode

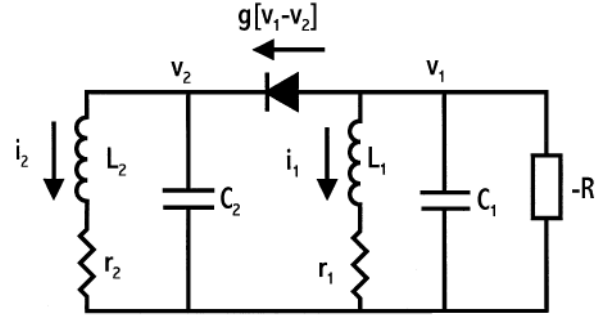


Fig. 1. Schematic diagram of a single hyperchaotic oscillator used in the experiments.

is represented by $g(v) = \rho I_d [\exp(\alpha v) - 1]$, where $I_d = 1.6$ nA and $\alpha = 22.6$ V⁻¹. We measure the current-voltage curve of the diode and fit it with the Shockley model $I = I_d [\exp(\alpha V) - 1]$ to obtain the value of the parameters α and I_d . The value of the reverse saturation current I_d is adjusted to improve the agreement between the experimentally observed and numerically generated phase space projections of the attractor at various values of a . With the chosen parameters, the model and the circuit display several periodic windows of similar structure, where the observed locations of these windows agree with the predictions of the model to within 1%.

The circuit displays fixed point, periodic, chaotic, or hyperchaotic dynamics depending on the value of the parameter a (set by the negative resistor). To characterize the various dynamic regimes, we determine the Lyapunov exponents of the circuit from experimental time series using the method of Eckmann *et al.* [1986]. We also compute the Lyapunov exponents of the circuit model and find reasonable agreement between the theoretical and experimental spectra, suggesting that the model closely resembles the experimental system. Figure 2 shows the two largest Lyapunov exponents in the neighborhood of the transition to hyperchaos. Hyperchaos is especially apparent beyond $a = 0.6$ where there are two large positive exponents. We perform our synchronization experiments with $a = 0.693$, well into the hyperchaotic regime. Figure 3 shows experimental phase space projections of the hyperchaotic attractor of the circuit for this value of a .

We attempt to synchronize the behavior of the two circuits using either of the two coupling schemes illustrated by a block diagram in Fig. 4. For each of these coupling schemes, there exists a range of coupling strength over which the largest transverse

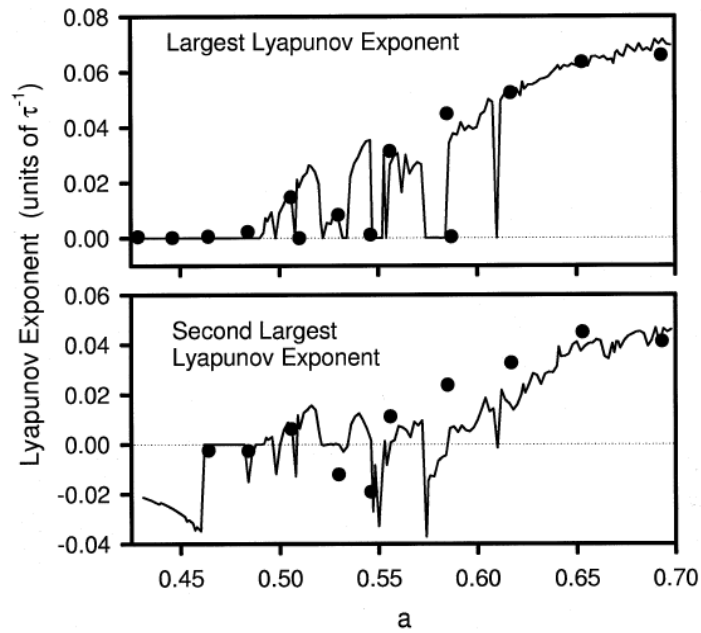


Fig. 2. The two largest Lyapunov exponents as a function of the bifurcation parameter a for a single hyperchaotic oscillator. The solid lines show the theoretically determined exponents and the solid circles show the exponents determined from experimental time series. The oscillator is hyperchaotic when both exponents are positive.

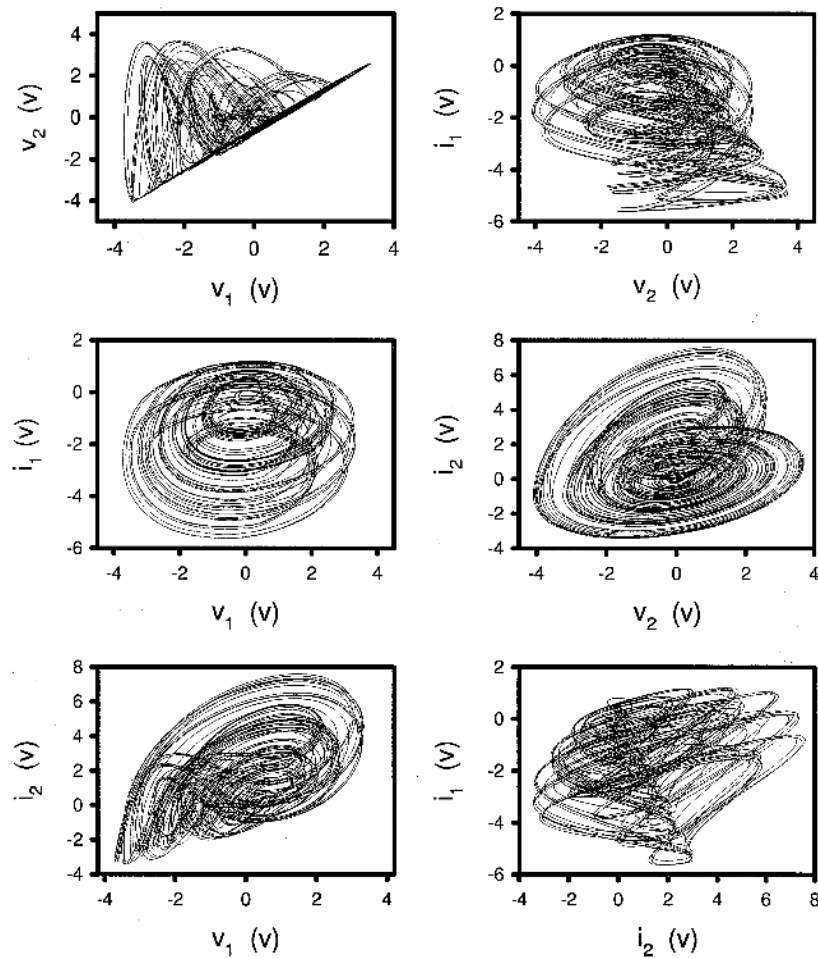


Fig. 3. Phase space projections of the hyperchaotic attractor of a single oscillator with $a = 0.693$.

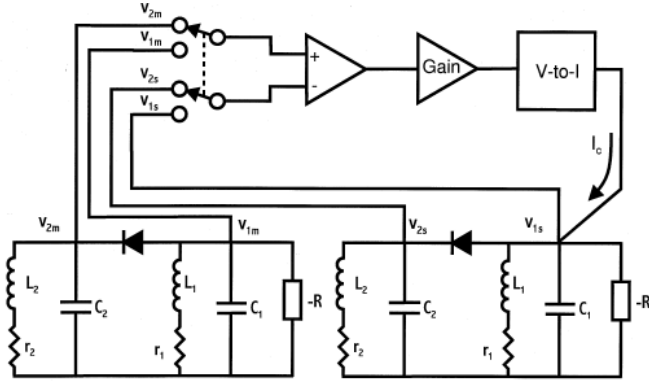


Fig. 4. Block diagram of coupling between master and slave circuits. The current I_c , proportional to the output of the differential amplifier, is injected into the slave circuit thereby coupling it to the master circuit. The coupling scheme is v_2 -to- v_1 (v_1) if the switch is up (down).

Lyapunov exponent is negative. In “ v_1 coupling” ($K_{11} = 1$, $K_{ij} = 0$ otherwise), the voltages across the capacitors C_1 of the master and slave are measured by high-impedance voltage followers (Texas Instruments TL072). The scaled difference of these

two signals is provided by an analog multiplier with differential inputs (Analog Devices AD633). The resulting voltage is further scaled by an inverting amplifier (OP-42) and converted to a current by a voltage-to-current converter based on an instrumentation amplifier (Analog Devices AD620) and operational amplifier (OP-42). This current is injected directly into the node above the capacitor C_1 of the slave circuit thereby coupling it to the master. Similarly, in “ v_2 -to- v_1 coupling” ($K_{21} = 1$, $K_{ij} = 0$ otherwise), the voltages across the capacitors C_2 of the master and slave circuits are measured by voltage followers, subtracted and processed by an analog multiplier and an inverting amplifier, converted to a current by an instrumentation amplifier, and injected into the node above the C_1 capacitor of the slave circuit. Note that this coupling scheme consists of measuring one variable (v_2) and feeding back to a different variable (v_1).

Analog electronics are also used to obtain a signal proportional to the square of the separation of the master and slave in phase space $|\mathbf{x}_\perp(t)|^2$ where, in the notation of Sec. 2,

$$\mathbf{x}_\perp = \mathbf{x}_m - \mathbf{x}_s = \begin{bmatrix} v_{1m} - v_{1s}, & v_{2m} - v_{2s}, & i_{1m} - i_{1s}, & i_{2m} - i_{2s} \end{bmatrix}^T. \quad (5)$$

As described in the previous paragraph, voltage followers are used to measure the voltages across the capacitors in each circuit. The currents flowing through the inductors are determined by using an instrumentation amplifier (AD620) to measure the voltage across a $1.96 \, \Omega$ resistor in series with each inductor. (In the model above, the resistors used to measure the currents are included in the values stated for r_1 and r_2 .) Analog multipliers with differential inputs provide the squares of the difference of these signals, which are then summed using an operational amplifier (OP42). The resulting signal is recorded and scaled by a computer using an analog-to-digital converter (National Instruments model AT-MIO-16H-9) to obtain $|\mathbf{x}_\perp(t)|^2$.

In our experiments, we quantify the degree of synchronization over a range of coupling strengths for each of the two coupling schemes described above. At each coupling strength, we record a time series of $|\mathbf{x}_\perp(t)|^2$ (where $|\bullet|$ denotes the Euclidean norm) of a duration on the order of $10^6\tau$. From the time series, we determine the maximum observed value $|\mathbf{x}_\perp(t)|_{\text{MAX}}^2$ and the root mean square average $|\mathbf{x}_\perp(t)|_{\text{RMS}}^2$, quantifying the occurrence of large separations between the master and slave

dynamics and the duration of these separations, respectively. For example, a large value of $|\mathbf{x}_\perp(t)|_{\text{MAX}}$ and a small value of $|\mathbf{x}_\perp(t)|_{\text{RMS}}$ indicate the presence of brief large desynchronization events in the time series, the hallmark of attractor bubbling.

4. Results

We first consider “ v_2 -to- v_1 coupling” ($K_{21} = 1$, $K_{ij} = 0$ otherwise) of the oscillators. Figure 5(a) shows the experimentally observed degree of synchronization, quantified by both the maximum observed value $|\mathbf{x}_\perp(t)|_{\text{MAX}}^2$ and the root mean square average $|\mathbf{x}_\perp(t)|_{\text{RMS}}^2$ as functions of the coupling strength c . Figure 5(b) shows the largest transverse Lyapunov exponent determined numerically from Eq. (3) using the method described by Jackson [1990]. The exponent is negative for coupling strengths in the range from $c = 0.66$ to $c = 1.62$, indicating that the synchronization manifold is asymptotically stable. Experimentally, we observe that $|\mathbf{x}_\perp(t)|_{\text{MAX}}^2$ is still comparable to the size of the attractor in this range, even though $|\mathbf{x}_\perp(t)|_{\text{RMS}}^2$ is very small, indicating the occurrence

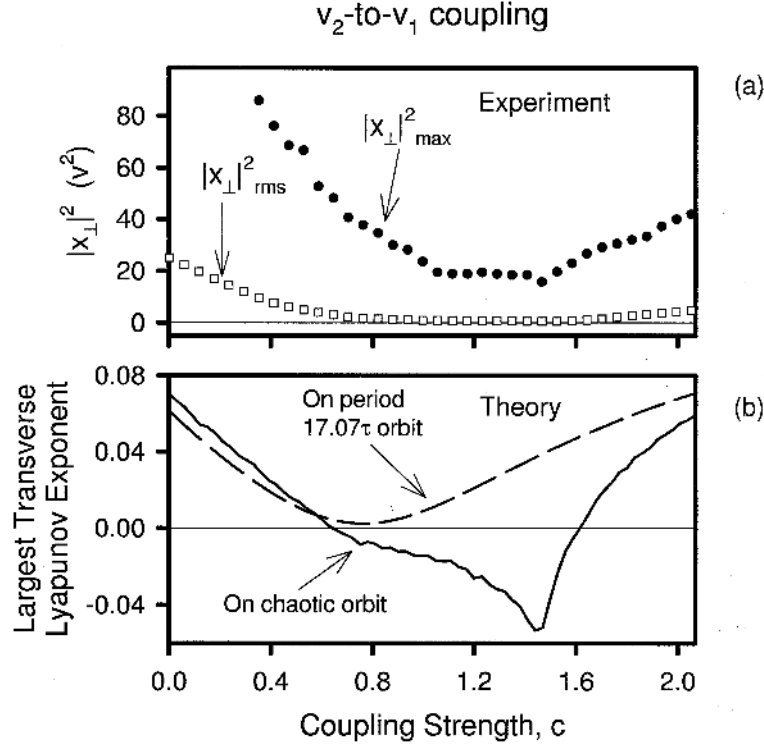


Fig. 5. (a) Experimentally observed degree of synchronization for v_2 -to- v_1 coupling. The squares indicate $|x_{\perp}|^2_{\text{RMS}}$ and the circles $|x_{\perp}|^2_{\text{MAX}}$. We observe desynchronization events at all coupling strengths as indicated by $|x_{\perp}| \gg 0$. (b) The largest transverse Lyapunov exponents as functions of coupling strength c calculated for both a chaotic trajectory (solid line) and an unstable periodic orbit embedded in the chaotic attractor (dashed line).

of attractor bubbling. With this coupling scheme, no high-quality synchronization is observed.

Next, we consider “ v_1 coupling” ($K_{11} = 1$, $K_{ij} = 0$ otherwise) of the oscillators. Figure 6 shows the experimentally observed degree of synchronization and the numerically determined largest transverse Lyapunov exponent. From Fig. 6(b), it is seen that the synchronization manifold is asymptotically stable for all coupling strengths larger than 0.26 since the exponent crosses zero at this point and remains negative for larger coupling strengths. Experimentally, we observe a small region of attractor bubbling occurring for the range of coupling strengths between 0.25 and 0.32. No desynchronization events greater than the noise level ($|x_{\perp}(t)|^2_{\text{MAX}} < 1.0 \text{ V}^2$ or 0.5% of the maximum possible value of $|x_{\perp}(t)|^2$ on the attractor) are observed for coupling strengths greater than 0.32. Thus, there is a large range of coupling strengths where high-quality synchronization can be achieved despite the hyperchaotic nature of the system. Figure 7 shows projections of the full eight-dimensional phase space of the coupled system onto the $v_{1m} - v_{1s}$ plane at three different coupling

strengths demonstrating the full range of observed behavior: Fig. 7(a) $c = 0$, no synchronization occurs; Fig. 7(b) $c = 0.28$, synchronization is degraded by attractor bubbling; Fig. 7(c) $c = 0.36$, high-quality synchronization is observed. Our observations are consistent with those of Tamaševičius *et al.* [1997b] who found that the transition to high-quality synchronization for this coupling scheme occurred for a coupling strength slightly higher than that expected based on the negativity of the transverse Lyapunov exponents.

Additional information concerning the bubbling transition can be obtained by observing the scaling of $|x_{\perp}(t)|^2_{\text{MAX}}$ in the vicinity of the transition. Venkataramani *et al.* [1996a] predict that the transition can be “hard” (the bursts appear abruptly with large amplitude as the coupling strength is varied) or “soft” (the maximum burst amplitude increases continuously from zero), and that the symmetry of the coupling has a fundamental effect on the transition. From Fig. 6(a), it is seen that the transition is soft. For such a transition and the one-way (asymmetric) coupling scheme used in our experiment, they predict

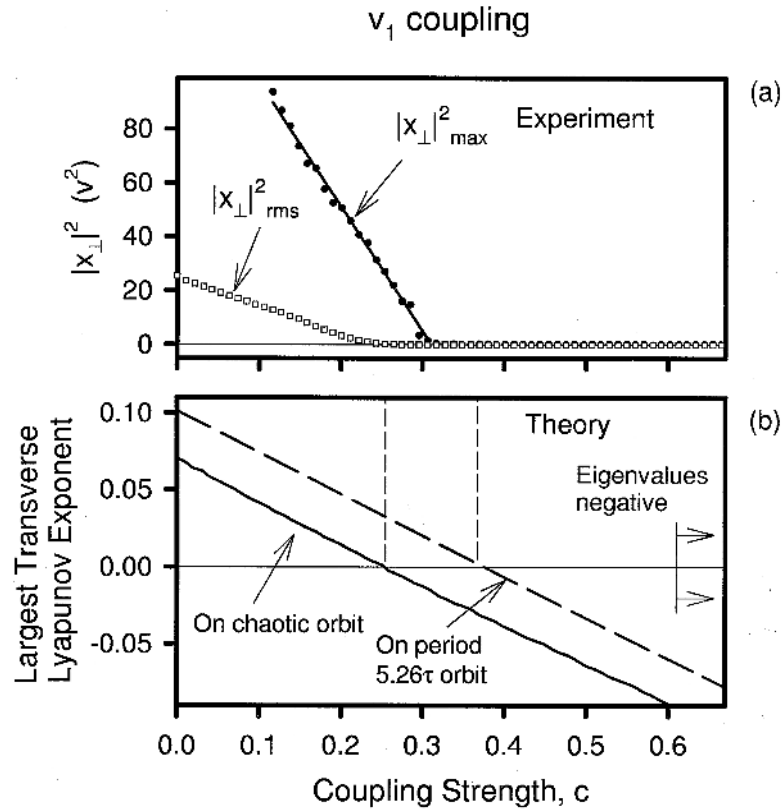


Fig. 6. (a) Experimentally observed degree of synchronization for v_1 coupling. The squares indicate $|\mathbf{x}_\perp|^2_{\text{RMS}}$ and the circles $|\mathbf{x}_\perp|^2_{\text{MAX}}$. We observe attractor bubbling in the range of coupling strengths $0.25 < c < 0.32$. The straight line indicates $|\mathbf{x}_\perp(t)|^2_{\text{MAX}} \propto (c_b - c)$ in the vicinity of the bubbling transition. (b) Largest transverse Lyapunov exponents as a function of coupling strength calculated for both a chaotic trajectory (solid line) and an unstable periodic orbit embedded in the chaotic attractor (dashed line). The vertical lines are a guide to the eye to facilitate the comparison between theory and experiment. The region in which the real parts of the eigenvalues of $\mathbf{J}$ are negative is indicated by the arrows on the right side of the figure.

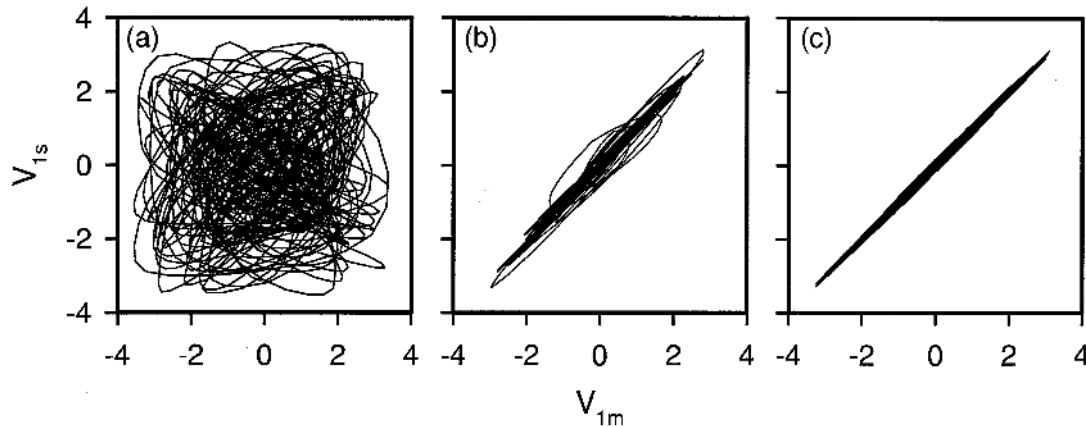


Fig. 7. Projection of the full eight-dimensional phase space of the v_1 coupled system onto the v_{1m} - v_{1s} plane at three different coupling strengths. (a) $c = 0$, $|\mathbf{x}_\perp(t)|^2_{\text{MAX}} = 254$ volts², $|\mathbf{x}_\perp(t)|^2_{\text{RMS}} = 25$ volts², no synchronization; (b) $c = 0.28$, $|\mathbf{x}_\perp(t)|^2_{\text{MAX}} = 16.1$ volts², $|\mathbf{x}_\perp(t)|^2_{\text{RMS}} = 1.4 \times 10^{-2}$ volts², synchronization is degraded by attractor bubbling; (c) $c = 0.36$, $|\mathbf{x}_\perp(t)|^2_{\text{MAX}} = 0.2$ volts², $|\mathbf{x}_\perp(t)|^2_{\text{RMS}} = 1.1 \times 10^{-2}$ volts², high-quality synchronization.

that $|\mathbf{x}_\perp(t)|^2_{\text{MAX}} \sim (c_b - c)^2$, where c_b is the coupling strength at which the transition occurs. On the contrary, we observe that $|\mathbf{x}_\perp(t)|^2_{\text{MAX}} \sim (c_b - c)$, as illustrated by the solid straight line shown in the

figure. The fit between our data and the straight line is good in that the deviation from the straight line is comparable to the observed datapoint-to-datapoint variation, which is a reasonable estimate

of our experimental error. Our observation indicates the existence of a new type of bubbling transition, or that the lowest-order nonlinear contribution to the transverse dynamics has an even symmetry even though the coupling has an odd symmetry.

5. Synchronization Criteria

The experiments described in the previous section, taken together with previous research on attractor bubbling, clearly demonstrate that the negativity of the largest transverse Lyapunov exponent is not a good predictor of high-quality synchronization in an experiment. Several alternative criteria for synchronization have been proposed, which make widely differing predictions for the range of coupling strength over which high-quality synchronization may be observed for our system. In this section, we review these proposed criteria and apply them to our experiment.

In the investigation of the stability of coupled oscillators, Ashwin *et al.* [1994] discovered the existence of attractor bubbling and found that it occurs when any unstable periodic set embedded in the chaotic attractor, such as an UPO, is characterized by a positive transverse Lyapunov exponent. Thus, a criterion for high-quality, bubble-free synchronization is that the largest transverse exponents characterizing each of the unstable sets embedded in the attractor must be negative. Since Lyapunov exponents are topological invariants, this criterion is independent of the choice of metric. Unfortunately, it is impossible to apply since there are typically an infinite number of unstable sets embedded in a chaotic attractor whose stability must be determined. There is some indication that the criterion might be applied in an approximate sense because it has been suggested that a low-period UPO typically yields the largest exponent [Hunt & Ott, 1996]. However, it is now known that this conjecture does not hold for an attractor near a crisis [Zoldi & Greenside, 1998; Hunt & Ott, 1998; Yang *et al.*, 1999].

To test this criterion, we have identified 11 of the low-period UPO's embedded in the hyperchaotic attractor using the method described in [Parker & Chua, 1989]. Our test is only approximate in the sense that we do not determine the stability of all UPO's. Note that $\mathbf{x}_{\parallel} = \mathbf{x}_{\perp} = 0$ is an unstable steady state of the system, but its neighborhood is never visited by the chaotic trajectory

and hence plays no role in attractor bubbling. For “ v_2 -to- v_1 ” coupling, the most transversely unstable set is a period-4 UPO (period of 17.07τ). The variation of the largest transverse Lyapunov exponent for this orbit as a function of coupling strength is shown in Fig. 5(b). This exponent is positive for all c . Thus, this criterion predicts attractor bubbling over the entire range where the largest transverse exponent of the chaotic attractor is negative ($0.66 < c < 1.62$). These predictions are consistent with our observations.

For v_1 coupling, the most transversely unstable set is a period-1 UPO (period of 5.26τ) whose largest transverse exponent as a function of coupling strength is shown in Fig. 6(b). Attractor bubbling is expected for coupling strengths in the range $0.25 < c < 0.37$ whereas it is observed experimentally in the smaller range $0.25 < c < 0.32$. Thus, this criterion overestimates the range of attractor bubbling, which may be due to the fact that many of the other UPO's become transversely stable around $c = 0.32$ or that the period-1 orbit is visited infrequently. Numerical simulations suggest that the orbit is visited very infrequently if at all. For our experiments, this criterion is somewhat successful at predicting attractor bubbling but its application suffers from uncertainty knowing whether enough unstable sets have been identified and whether the attractor visits the neighborhood of each set.

Based on the same idea that the stability of the unstable sets govern the region of attractor bubbling, Brown and Rulkov [1997] suggest an alternative method for determining the stability of these sets that is easier to implement. They derived a sufficient, but not necessary, condition for the asymptotic stability using Gronwald's theorem. Briefly, they decompose the matrix $\mathbf{J} = D\mathbf{F}[\mathbf{x}_m(t)] - \gamma\mathbf{K}$ into a time-independent part $\mathbf{A} \equiv \langle D\mathbf{F} \rangle - \gamma\mathbf{K}$ and a time-dependent part $\mathbf{B}(\mathbf{x}_m; t) \equiv D\mathbf{F}[\mathbf{x}_m(t)] - \langle D\mathbf{F} \rangle$, where $\langle \bullet \rangle$ denotes a time average over the driving trajectory. A trajectory is transversely stable when

$$-\text{Re}[\Lambda_1] > \langle \|\mathbf{P}^{-1}[\mathbf{B}(\mathbf{x}_m; t)]\mathbf{P}\| \rangle, \quad (6)$$

where Λ_1 is the largest eigenvalue of $\mathbf{A}$, $\mathbf{P}$ is a matrix of eigenvectors of $\mathbf{A}$, and $\|\bullet\|$ denotes a norm whose choice is arbitrary. As for the previous criterion, this condition must be evaluated along all unstable sets embedded in the attractor to determine the range of coupling strengths over which attractor bubbling can be avoided. Again,

this task is not trivial because of the infinite number of unstable sets. In addition, the predictions of this criterion depend on both the choice of a norm and the metric and hence it is not topologically invariant.

We applied this criterion to our experiment by evaluating the stability of the 11 lowest period UPO's. For both coupling schemes and all coupling strengths, the right-hand side of Eq. 6 is always extremely large and the condition is never satisfied. Hence, this criterion correctly predicts that high-quality synchronization does not occur for v_2 -to- v_1 coupling. On the other hand, it predicts that high-quality synchronization cannot be achieved with v_1 coupling, whereas our experiments show a large regime of bubble-free synchronization. Apparently this sufficient, but not necessary, criterion is overly conservative in this case.

To avoid the impracticality of applying the criteria described above, two others have been suggested. Pecora *et al.* [1995] attempt to guarantee high-quality synchronization by requiring all eigenvalues of the matrix $\mathbf{J}$ have negative real parts at all points along the chaotic driving trajectory $\mathbf{x}_m(t)$. The justification for this condition has some motivation from the control literature [Brogan, 1974], even though the eigenvalues of a time-varying linear system cannot be used to determine the asymptotic behavior of the solutions in general [Hale, 1969]. As for the previous case, this criterion depends on the choice of metric.

Applying this criterion to our experiment by determining numerically the eigenvalues of $\mathbf{J}$ along a chaotic trajectory of duration $10^6\tau$, we find that there are always points where the real part of an eigenvalue is positive for v_2 -to- v_1 coupling. Thus, no high-quality synchronization is predicted for this coupling scheme, consistent with our experimental observations. For v_1 coupling, all eigenvalues have negative real parts for $c > 0.61$. Thus, the predicted range of high-quality synchronization is smaller than that observed in our experiments. This criterion provides a reasonable estimate of the regime of high-quality synchronization without the impracticality of determining the transverse stability of an infinite number of unstable sets. We note, however, that this criterion has failed in other experiments [Blakely *et al.*, 2000].

A second criterion was suggested by Gauthier and Bienfang [1996]. They argued that the effect of noise in attractor bubbling is to repeatedly set the system into a transient behavior where a

perturbation grows rapidly for a brief time before decaying. Using this reasoning, they proposed a method for testing whether perturbations undergo transient growth. This criterion is based on the time derivative of the Lyapunov function $\mathcal{L} \equiv |\delta\mathbf{x}_\perp(t)|^2$. A sufficient condition that all perturbations decay to the manifold without transient growth is

$$\frac{d\mathcal{L}}{dt} = 2\delta\mathbf{x}_\perp(t) \cdot \mathbf{J}\delta\mathbf{x}_\perp(t) < 0 \quad (7)$$

for all times. As in the two previous cases, this criterion depends on the choice of metric and norm.

For our experiment, Eq. (7) is simple enough to be evaluated analytically. For the v_2 -to- v_1 coupled oscillators,

$$\begin{aligned} \frac{d\mathcal{L}}{dt} = & (a - g')\delta x_1^2 - h\delta x_3^2 - e \left(\delta x_4 - \frac{(d-b)}{2e}\delta x_2 \right)^2 \\ & - [(b+1)g' - c]\delta x_1\delta x_2 \\ & + \left(\frac{(d-b)^2}{4e} - bg' \right) \delta x_2^2, \end{aligned} \quad (8)$$

where $\delta\mathbf{x}_\perp = [\delta x_1, \delta x_2, \delta x_3, \delta x_4]^T$, and $g'[v] = \alpha\rho I_d \exp(\alpha v)$. We see that $d\mathcal{L}/dt$ can be positive regardless of the value of the coupling strength c by considering the case when $\delta x_2 = \delta x_3 = \delta x_4 = 0$, and $\delta x_1 \neq 0$. In this situation, only the first term is nonzero and can be positive since g' is near zero for some part of the trajectory. Thus, the criterion predicts no high-quality synchronization, consistent with our experimental observations.

For the v_1 coupled oscillators,

$$\begin{aligned} \frac{d\mathcal{L}}{dt} = & (a - g' - c)\delta x_1^2 - h\delta x_3^2 \\ & - e \left(\delta x_4 - \frac{(d-b)}{2e}\delta x_2 \right)^2 - (b+1)g'\delta x_1\delta x_2 \\ & + \left(\frac{(d-b)^2}{4e} - bg' \right) \delta x_2^2. \end{aligned} \quad (9)$$

We see that $d\mathcal{L}/dt$ can be positive regardless of the value of the coupling strength c by considering the case when $\delta x_1 = \delta x_3 = 0$, and $\delta x_4 = (d-b)\delta x_2/2e \neq 0$. In this situation, only the last term remains and can be positive since g' is near zero for some part of the trajectory. Thus, no high-quality synchronization should be expected. In fact, a large range of high-quality synchronization is observed in the experiment and hence this criterion is much too conservative.

6. Discussion

In summary, we investigate attractor bubbling in an experimental system of coupled hyperchaotic oscillators. Our results show clearly that attractor bubbling is present over all coupling strengths for one coupling scheme and hence high-quality synchronization is not possible for this scheme. On the other hand, a large regime of high-quality synchronization is observed for a second coupling scheme. In this case, we measure the scaling of the square of the burst amplitude as a function of coupling strength near the bubbling transition point. The scaling does not fit into the known categories, potentially indicating a new type of transition. We attempt to predict the regime of high-quality synchronization using several proposed synchronization criteria. The criterion proposed by Ashwin *et al.* [1994] appears to provide a reasonable estimate of the range of high-quality synchronization but it can only be implemented approximately because it requires analysis of an infinite number of unstable sets. The criterion proposed by Pecora *et al.* [1995] provides the next best estimate and is much simpler to apply. However, it has been shown to fail in other experiments [Blakely *et al.*, 2000]. The criteria proposed by Brown and Rulkov [1997] and Gauthier and Bienfang [1996] are both too conservative for our experimental system.

Our results have clear implications for secure communications using synchronized chaos. Communication systems based on low-dimensional chaos have proven to be susceptible to eavesdropping methods based on nonlinear prediction techniques [Short, 1997] or even simple return maps [Perez & Cerdeira, 1995]. To improve security, it has been suggested that synchronized hyperchaos provides more complex signal masking [Peng *et al.*, 1996]. However, faithful recovery of a masked message requires high-quality synchronization. We find that attractor bubbling does occur over a wide range of coupling strengths for some coupling schemes but it is not highly prevalent for other schemes. Therefore, attractor bubbling may not hinder the design of communication systems based on hyperchaotic oscillators.

Unfortunately, our comparison of the proposed synchronization criteria indicates that there is no simple and generally reliable way to determine how to couple two oscillators to achieve high-quality synchronization. Additional research is needed to address this issue.

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Appendix A

Determining the Lyapunov Exponents

We determine the Lyapunov exponents of our oscillators from experimental time series using the method introduced by Eckmann *et al.* [1986] and extended by Brown *et al.* [1990]. Briefly, this method consists of the following steps: (1) for each data point x_i , search the time series to obtain a ball of points in a small neighborhood of x_i ; (2) approximate the evolution of each neighborhood one time step forward with a linear map; (3) determine the eigenvalues of a product of the matrices using the

QR decomposition to obtain the Lyapunov exponents. Note that all four variables are directly accessible in our experiment and hence no time-delay phase space reconstruction is necessary. We measure all four variables and represent the range of voltages from -2.82 volts to 2.82 volts in integer format using 65536 integers. Each time series consisted of 200,000 points with a sample interval of $0.21 \mu\text{sec}$. At each point in the time series we search for 32 neighbors. Following Brown *et al.* [1990], we use a Hanning window to reduce end effects.

We use a similar procedure to determine the exponents of the circuit model. Rather than searching a time series for near neighbors, we generate random nearby points in phase space and evolve them forward using the equations of the model. This allows us to obtain arbitrarily small and well-populated neighborhoods. The values reported in Sec. 3 are obtained using 16 point neighborhoods within a distance of 10^{-5} V. The length of the model time series is 200,000 points.